

AMR BELAL ARAB

Applied AI & Generative AI Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Applied AI and Generative AI Engineer building end-to-end intelligent products across LLM applications, agentic and multi-agent workflows, NLP, computer vision, and machine learning. Experienced in connecting models to user-facing applications through Python APIs, structured workflows, automated testing, and deployment, with a focus on reliable behavior and real-world utility.

EDUCATION

Helwan International Technological University (HITU)

2022 - 2026

B.Sc. Artificial Intelligence | Excellent Honors | CGPA eq. 3.57/4.00 (89.21%) | Cohort Rank: 10th

Relevant coursework: Machine Learning, Neural Networks, Computer Vision, NLP, Data Structures, Databases, Software Engineering.

EXPERIENCE

Freelance AI Developer

Mar 2024 - Present

- Designed Python-based AI solutions across NLP, clustering, anomaly and outlier detection, and data analysis, translating client requirements into working prototypes and applications.
- Built data preparation, model training, evaluation, visualization, and lightweight web-delivery workflows using scikit-learn, TensorFlow, Flask, Streamlit, and related tooling.

SELECTED PROJECTS

Weekly GenAI LinkedIn Agent

Jul 2026

Personal multi-agent project | [GitHub](#)

- Built an automated multi-agent workflow that researches current Generative AI news, cross-checks stories across sources, ranks updates, and drafts professional Arabic and English LinkedIn posts.
- Added structured outputs, deduplication, an automated quality gate, email delivery, scheduling with GitHub Actions, and Pytest coverage for a repeatable weekly pipeline.

SourceCraft Research - Multi-Agent Research Assistant

Jul 2026

[GitHub](#) | [Live Demo](#)

- Designed a bounded three-agent LangGraph workflow that searches the web, writes cited Markdown briefs, reviews its own output, and routes evidence or writing gaps back to the appropriate agent.
- Packaged the system as a Streamlit application with structured state, Gemini API integration, DuckDuckGo search, Pytest, Ruff, Docker, and CI.

ISH - Intelligent Spherical Home Robot

2025 - 2026

College graduation project | [GitHub](#) | [Documentation](#)

- Built an AI-powered spherical smart-home robot that combines assistance, mobile security, environmental safety monitoring, reminders, object-location memory, and remote access in one modular system.
- Integrated face recognition, Arabic and English voice interaction, local TTS, ESP32-CAM streaming, sensor alerts, FastAPI services, a Flutter app, Supabase, ESP32, and Raspberry Pi components.

Raay - Amazon Egypt Sentiment & AI Marketing

Jul 2026

DEPI graduation project - Generative AI Track, Round 4 | [GitHub](#) | [Live Demo](#)

- Developed a Streamlit application that searches Amazon Egypt, collects Arabic and English reviews, analyzes sentiment, compares products, visualizes insights, and exports results.
- Added automated generation of a social-ready marketing poster for the strongest product using Transformers, Hugging Face tooling, Stable Diffusion, Pillow, and tested data workflows.

SELECTED PROJECTS - CONTINUED

Student Course Registration System - HCI Prototype May 2026
[GitHub](#)

- Created a three-tier Flask and SQLite prototype with authentication, dashboards, CRUD workflows, advanced search, validation, and HCI-driven error prevention and feedback.

FireVision - Fire Image Classifier Mar 2026
[GitHub](#)

- Evaluated 88 deep-learning experiments for fire versus non-fire classification across original images, augmentation, seven edge-detection methods, and combined preprocessing pipelines.
- Achieved 97.96% accuracy with EfficientNetB0 using rotation or contrast augmentation, demonstrating that augmentation outperformed edge-based preprocessing for this dataset.

Information Retrieval Engine & Flask Interface Nov 2025
[GitHub](#)

- Built and evaluated a TF-IDF retrieval pipeline over 2,000 movie reviews using NLTK and Gensim, then deployed a Flask search interface with top-k results and precision, recall, and F1 evaluation.

IoT Weather Prediction & Smart Control System Nov 2025
[GitHub](#)

- Connected ESP32 sensors for temperature, humidity, rain, wind, and light to an ML decision pipeline and real-time dashboard with switchable manual and AI control modes.

TRAINING & CERTIFICATIONS

Generative AI Intern - DEPI / EYouth Nov 2025 - Jul 2026

- Deep learning, Transformers, LLMs, Hugging Face, GANs, diffusion models, Agentic AI, and multi-agent systems.

Data Science Intern - DEPI / AMIT Oct 2024 - May 2025

- Data cleaning, EDA, feature engineering, statistics, and model evaluation.

AI Intern - The British University in Egypt Feb 2025

- Computer vision, CNNs, transfer learning, and YOLO fundamentals.

AI & ML Training - Shabab Mubtakirun / Huawei Jan 2025 - Feb 2025

- Machine learning pipelines, regression, classification, clustering, and preprocessing.

TECHNICAL SKILLS

Applied & Generative AI: LLM applications, Agentic AI, multi-agent systems, LangGraph, Gemini API, prompt engineering, structured outputs, automated quality gates

Machine Learning: scikit-learn, TensorFlow, Keras, Transformers, NLP, computer vision, classification, clustering, model evaluation

Engineering & Delivery: Python, FastAPI, Flask, Streamlit, REST APIs, Docker, GitHub Actions, Pytest, Ruff, Git

Data & Systems: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Supabase, PostgreSQL, ESP32, Raspberry Pi, IoT